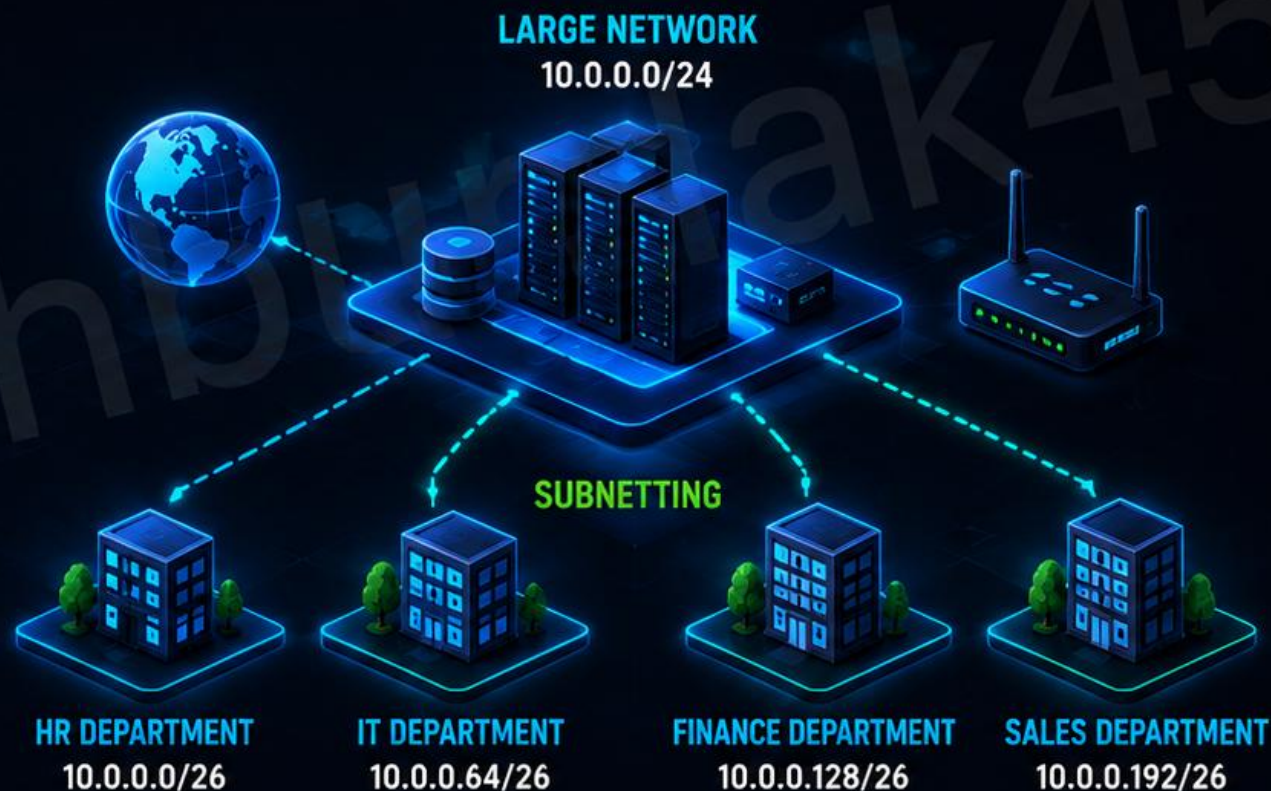


SUBNETTING & NETWORK LAYER

Simplified From Zero



Subnetting is the process of dividing a large network into smaller, efficient sub-networks for better performance, security and management.



EFFICIENT IP MANAGEMENT

Use IP addresses wisely and effectively.



IMPROVED SECURITY

Isolate networks and limit unauthorized access.



BETTER PERFORMANCE

Reduce broadcast traffic and network congestion.



EASY NETWORK SCALABILITY

Grow your network without complexity.



EASY TROUBLESHOOTING

Identify and resolve issues quickly.



Understand once, apply forever – **Subnetting** is the foundation of every network you build.

02

WHY NETWORKING MODELS & SUBNETTING MATTER

CONNECT. COMMUNICATE. COLLABORATE.
THE WORLD RUNS ON NETWORKS.



Networking models give structure to complex communication, while **subnetting** helps break big networks into smaller, smarter, and more secure parts.

They bring order to chaos.



THEY HELP US TO:



IMPROVE SECURITY

Isolate departments and systems to prevent unauthorized access and attacks.



BOOST PERFORMANCE

Reduce network congestion and improve data flow and speed.



ENHANCE SCALABILITY

Easily grow your network without disrupting the whole system.



SIMPLIFY MANAGEMENT

Smaller subnets are easier to manage, monitor, and troubleshoot.



SAVE RESOURCES

Efficient IP address usage means less waste and optimized cost.



REAL WORLD IMPACT

Every organization – from startups to tech giants – relies on subnetting to keep their networks efficient, secure, and future-ready.

STARTUP



10.1.1.0/24

ENTERPRISE



10.1.2.0/24

CAMPUS NETWORK



10.1.3.0/24

CLOUD INFRASTRUCTURE



10.1.4.0/24

“ A WELL DESIGNED NETWORK ISN'T JUST ABOUT CONNECTING DEVICES, IT'S ABOUT BUILDING A **SECURE, SMART, AND SCALABLE** FOUNDATION. ”

03

OSI MODEL OVERVIEW

The 7-Layer Framework That Powers Every Connection

“Understanding the OSI model is the first step to mastering networks.”



THE 7 LAYERS OF OSI MODEL

7		APPLICATION LAYER	Network services to applications (HTTP, FTP, DNS, SMTP)
6		PRESENTATION LAYER	Data format, encryption, compression, translation
5		SESSION LAYER	Manages sessions between applications
4		TRANSPORT LAYER	End-to-end communication, reliability, flow control (TCP, UDP)
3		NETWORK LAYER	Logical addressing, routing, path selection (IP, ICMP)
2		DATA LINK LAYER	Node-to-node delivery, framing, MAC addressing (Switch)
1		PHYSICAL LAYER	Transmits raw bits over physical media (Cables, Signals)

SPOTLIGHT ON LAYER 3 – THE NETWORK LAYER

The **Network Layer** is responsible for moving packets from the source to the destination across multiple networks. It uses **logical IP addresses** to find the best path and ensures the data reaches the correct network.



HOW LAYER 3 WORKS (ROUTING IN ACTION)



KEY TAKEAWAY



Layer 3 makes sure your data takes the right path to reach its destination, no matter how far it is.

COMMON LAYER 3 PROTOCOLS



IP
Internet Protocol



ICMP
Error & Control Messages



ARP
IP to MAC Mapping



RIP / OSPF / BGP
Routing Protocols

IN REAL LIFE



Routers work at Layer 3. Switches work at Layer 2. Your data moves up and down the OSI stack to communicate.

REMEMBER

No Layer 3,
No Routing.
No Routing,
No Internet!

4

IPV4 ADDRESS – THE BASICS

Every device on a network needs a **unique address** to communicate.

IPV4 ADDRESS FORMAT

An IPv4 address is a **32-bit** address, written in dotted-decimal notation.



EXAMPLE IN REAL LIFE



Your Device
192.168.1.10



Router
192.168.1.1



Internet



Just like your home address helps the postman deliver letters, IP address helps data reach the **right device**.

IPV4 ADDRESS CLASSES (BY FIRST OCTET)

Class	First Octet Range	Default Subnet Mask	Use
A	1 – 126	255.0.0.0	Very Large Networks
B	128 – 191	255.255.0.0	Medium Networks
C	192 – 223	255.255.255.0	Small Networks
D	224 – 239	N/A (Multicast)	Multicasting
E	240 – 255	N/A (Reserved)	Research/Reserved

KEY POINTS



IPv4 uses
32 bits



Written in
dotted-decimal
(4 numbers)



Divided into
network part
and **host** part



Helps in
identifying **networks**
and **devices**



ONE SIMPLE RULE: Every device, every connection, every communication starts with an **IP address!**



5

PRIVATE vs PUBLIC IP ADDRESS

Some IP addresses are **private** (for internal use), others are **public** (for the internet).

“ **Private IP** → Stay inside.
Public IP → Go worldwide. ”

PRIVATE IP ADDRESS

Used inside local networks (home, office, big companies).

Not reachable directly from the internet.

PRIVATE RANGE	SUBNET MASK	USABLE IPs	CLASS
10.0.0.0 – 10.255.255.255	255.0.0.0	16,777,214	A
172.16.0.0 – 172.31.255.255	255.240.0.0	1,048,574	B
192.168.0.0 – 192.168.255.255	255.255.0.0	65,534	C

EXAMPLE (Home Network)



Private IPs save public IPs and keep internal networks **secure**.

PUBLIC IP ADDRESS

Used on the internet to identify devices worldwide.

Reachable from **anywhere** on the internet.



Unique
World-wide
unique



Internet
Used to access
the internet



Provided by
ISP (Internet
Service Provider)



Used by
Websites, Servers,
Organizations

EXAMPLE (Web Server)



Web Server
203.0.113.5



Users from
anywhere

QUICK COMPARISON

- Used inside local networks
- Not directly reachable from internet
- Reusable in any organization

PRIVATE IP



VS

PUBLIC IP



- Used on the internet
- Directly reachable from internet
- Globally unique



KEY TAKEAWAY

Private IP = Internal Communication
Public IP = Global Communication

6

IPV4 ADDRESS vs MAC ADDRESS

Two different addresses. Two different jobs.

IP ADDRESS (Layer 3 – Network Layer)

Logical address used to identify a device **across networks**.



Logical Address

Assigned by the network (manually or via DHCP).



Used for Routing

Helps packets travel from source to destination.



Example

192.168.1.10



Works at Layer 3

Network Layer of OSI Model.

ACROSS DIFFERENT NETWORKS



Think of **IP** as your **HOME ADDRESS** (City + House).



Tells the postman (router) which city and house to go to.

MAC ADDRESS (Layer 2 – Data Link Layer)

Physical address used to identify a device **within the local network**.



Physical Address

Burned into the network interface card (NIC).



Used for Local Delivery

Helps frames move within the same local network.



Example

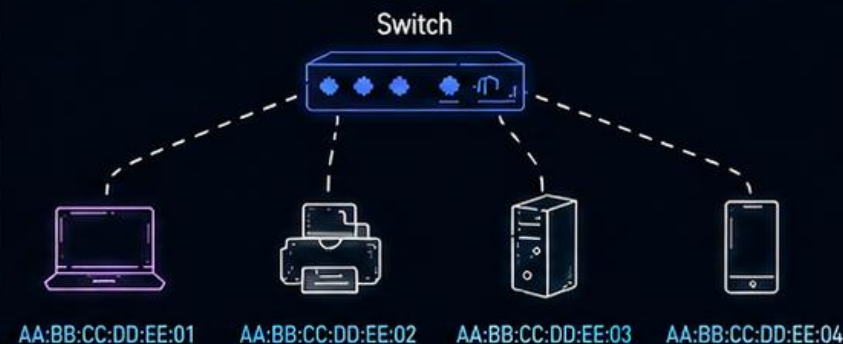
AA:BB:CC:DD:EE:FF



Works at Layer 2

Data Link Layer of OSI Model.

WITHIN THE SAME LOCAL NETWORK



Think of **MAC** as your **APARTMENT NUMBER** (Flat).



Tells the postman (switch) which flat/door to deliver inside.

KEY DIFFERENCES

Feature	IP Address	MAC Address
Purpose	Identifies device across networks	Identifies device within local network
Layer	Network Layer (Layer 3)	Data Link Layer (Layer 2)
Changeable?	Yes (Can be changed)	No (Usually permanent)
Length	32 bits (IPv4)	48 bits
Example	192.168.1.10	AA:BB:CC:DD:EE:FF
Used by	Routers	Switches

HOW THEY WORK TOGETHER



KEY TAKEAWAY:

IP address is for the **journey**, MAC address is for the **next door**.



Different roles. Same mission – Successful Communication.

7

CIDR NOTATION – IP MADE SIMPLE

CIDR (Classless Inter-Domain Routing) makes IP addressing **smarter** and **flexible**.



CIDR removes the class concept and **saves IP addresses!**



WHAT IS CIDR?

CIDR lets us write an IP address and its network size in one line using a **prefix**.

Format: **IP Address / Prefix Length**

EXAMPLE

192.168.1.0 / 24

IP Address

Network
Address

Prefix Length

Number of bits
in the network part

HOW CIDR WORKS

The prefix length tells how many leading bits are for the **network**.



192.168.1.0 / 24

1 = Network Bit

0 = Host Bit

/24 means 24 bits are for the network.



Efficient

Uses IP addresses more efficiently.



Scalable

Supports variable network sizes.



Flexible Routing

Helps in route summarization.



Easy to Manage

Simplifies IP allocation and management.

COMMON CIDR PREFIXES

Prefix	Subnet Mask	Network Bits	Host Bits	Usable Hosts
/8	255.0.0.0	8	24	16,777,214
/16	255.255.0.0	16	16	65,534
/24	255.255.255.0	24	8	254
/25	255.255.255.128	25	7	126
/26	255.255.255.192	26	6	62
/30	255.255.255.252	30	2	2

VISUAL UNDERSTANDING

IP / Prefix

Network Part

Host Part

Usable Hosts

/8

11111111

.....

00000000

00000000

00000000

16,777,214

/16

11111111

11111111

.....

00000000

00000000

65,534

/24

11111111

11111111

11111111

.....

00000000

254

/30

11111111

11111111

11111111

11111100

.....

00

2



KEY TAKEAWAY:

CIDR gives us the power to use IP addresses **wisely** and build better, **scalable** networks.



CIDR = **Smart IP, Smart Network!**

8

SUBNETTING – BREAK BIG, BUILD SMART

Subnetting divides a big network into smaller parts for **better**, **smarter**, **safer** networks.

“ Subnetting today,
better networks **tomorrow**. ”

WHAT IS SUBNETTING?

Subnetting means dividing a large network into smaller networks (**subnets**).

WHY SUBNET?



Better Performance

Reduces network congestion.



Improved Security

Limits access within a subnet.



Efficient Management

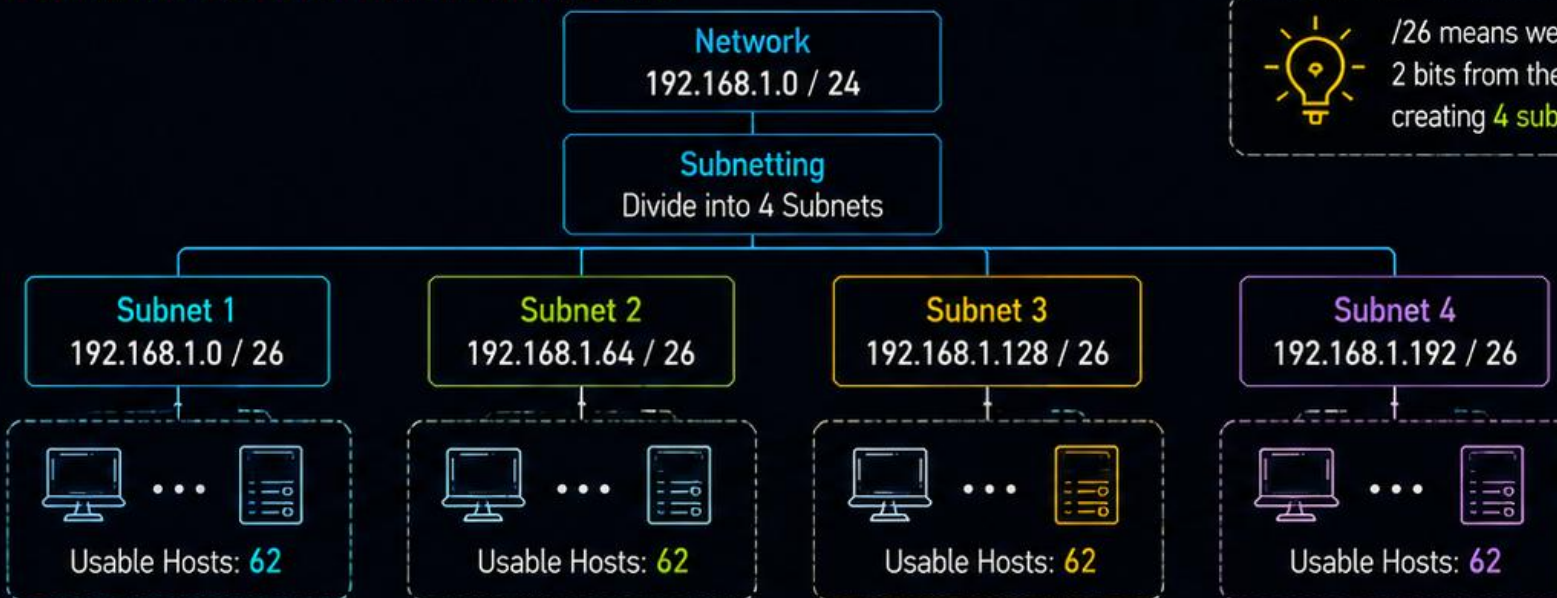
Easier to manage and troubleshoot.



IP Address Saving

Uses IP addresses efficiently.

EXAMPLE: SUBNETTING A NETWORK



/26 means we borrowed
2 bits from the host part,
creating **4 subnets**.

HOW SUBNETTING WORKS (IN SIMPLE STEPS)

①



Take a big
network
(e.g. /24)



②



Borrow bits from
the **host part**.



③



Create multiple
subnets.



④



Assign devices
to **subnets**.



⑤



Enjoy a **better**,
safer network!

SUBNETTING CALCULATION (EXAMPLE)

Details	/24 (No Subnetting)	/26 (After Subnetting)
Network	192.168.1.0	192.168.1.0, 64, 128, 192
Subnet Mask	255.255.255.0	255.255.255.192
Subnets	1	4
Usable Hosts per Subnet	254	62
Total Usable Hosts	254	248 (4 × 62)



KEY TAKEAWAY:

Subnetting breaks a big network into smaller pieces for **performance**, **security**, and **efficiency**.



Small steps. **Better networks**.

9

BROADCAST ADDRESS – REACH EVERYONE

Broadcast address is used to send data to **all devices** in a network.



One address.

One message. **Everyone** receives.



WHAT IS BROADCAST ADDRESS?

The highest IP address in a network.
Used to send data to **all devices** in that network.



Reaches Everyone

Data sent to broadcast address is received by **all devices** in the network.



Important for Communication

Used in network discovery, updates, and announcements.



Not Routed

Broadcasts stay within the local network. Routers do not forward them.



Useful Protocols

Used by DHCP, ARP, and network tools for smooth communication.

HOW BROADCAST ADDRESS IS FORMED

Set all host bits to 1.

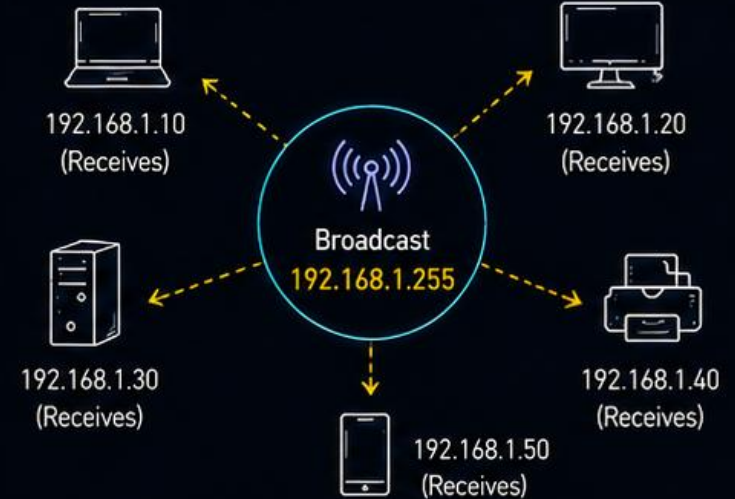
	Network Part (Remains Same)			Host Part (All 1s)
IP Address	11000000	10101000	00000001	00000000
Broadcast Address	11000000	10101000	00000001	11111111

EXAMPLE

Network: 192.168.1.0 / 24

IP Address (Lowest)	192.168.1.0
Broadcast Address (Highest)	192.168.1.255
Usable IP Range	192.168.1.1 – 192.168.1.254

BROADCAST IN ACTION



Data sent to broadcast address is delivered to every device in the same network.

BROADCAST ADDRESS CHEAT SHEET

Item	Details
Definition	Highest IP address in a network (all host bits = 1)
Purpose	Send data to all devices in the network
Format	Same network part, all 1s in host part
Routed?	No, broadcast is not routed
Example (/24)	Network: 192.168.1.0 Broadcast: 192.168.1.255
Protocols Using It	DHCP, ARP, NetBIOS, Routing Updates

COMMON SUBNETS – BROADCAST ADDRESSES

Network (CIDR)	Subnet Mask	Broadcast Address
192.168.1.0 / 24	255.255.255.0	192.168.1.255
192.168.1.0 / 25	255.255.255.128	192.168.1.127
192.168.1.128 / 25	255.255.255.128	192.168.1.255
10.0.0.0 / 24	255.255.255.0	10.0.0.255

KEY TAKEAWAY



- Broadcast address = **All 1s** in host part.
- Reaches **all devices** in the same network.
- Useful, but too much broadcast can cause **network congestion**.



REMEMBER:

Unicast is for **one**. Broadcast is for **all**. Use broadcast **wisely**!



Smart networks use the right address for the **right job**.

10

STATIC IP vs DYNAMIC IP – FIXED or FLEXIBLE?

Two ways to get an IP address. Two different uses. Both are **important!**

“ Static stays the same.
Dynamic keeps things running. ”

FEATURE	STATIC IP ADDRESS	DYNAMIC IP ADDRESS
 Definition	A fixed IP address that does not change.	An IP address that changes automatically.
 Changes?	Never changes (unless manually changed).	Changes from time to time .
 Assigned By	Manually configured by user or admin.	Automatically assigned by DHCP .
 Control	Full control over the IP address.	Limited control (assigned by ISP/Router).
 Use Cases	Servers, websites, CCTV, remote access.	Home users, mobile devices, offices.
 Cost	Usually requires extra payment.	Usually free (default option).
 Example	203.0.113.10	192.168.1.105 (today) 192.168.1.106 (tomorrow)

WHERE ARE THEY USED?

STATIC IP – Best For



Web
Servers



IP
Cameras



Remote
Access



Email
Servers

DYNAMIC IP – Best For



Home
Internet



Mobile
Devices



Office
Networks



Guest
Networks

HOW IT LOOKS IN REAL LIFE

STATIC IP (Fixed)



DYNAMIC IP (Changes)



QUICK COMPARISON

	STATIC IP	DYNAMIC IP
 Consistency	Always Same	Changes
 Reliability	Always Same	Changes
 Cost	High (Stable)	Medium (Varies)
 Best For	Paid	Free
	Servers & Critical Use	General Use



KEY TAKEAWAY:

Static IP gives **stability** and **control**.

Dynamic IP gives **flexibility** and **convenience**.



Choose what fits your **need!**

11

SUBNET MASK – THE NETWORK FILTER

Subnet mask helps separate **network part** and **host part** of an IP address.

“ Subnet mask is the **filter** that divides **network** and **host**. ”

WHAT IS SUBNET MASK?

A subnet mask is a 32-bit number that divides an IP address into two parts:



Network Part – Identifies the network.



Host Part – Identifies the device within that network.

COMMON SUBNET MASKS

CIDR	Subnet Mask	Usable Hosts
/8	255.0.0.0	16,777,214
/16	255.255.0.0	65,534
/24	255.255.255.0	254
/25	255.255.255.128	126
/26	255.255.255.192	62
/30	255.255.255.252	2

HOW SUBNET MASK WORKS

■ 1 = Network Bit ■ 0 = Host Bit

	Network Part												Host Part				
	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0
	Network Part												Host Part				
IP Address	192			.	168			.	1			.	10				
Subnet Mask	255			.	255			.	255			.	0				
Binary	11000000			.	10101000			.	00000001			.	00001010				
Result (Network ID)	192			.	168			.	1			.	0				



Subnet mask is used in IP address calculation, routing, and subnetting.

TYPES OF SUBNET MASKS

1. Default (Classful)



Class	Default Mask	CIDR
A	255.0.0.0	/8
B	255.255.0.0	/16
C	255.255.255.0	/24

2. Custom (CIDR)



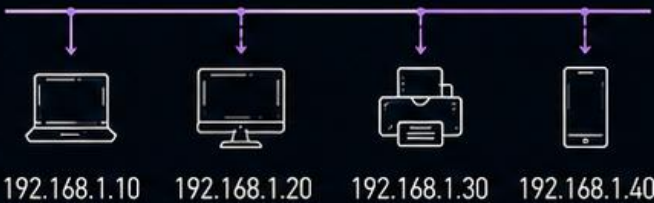
Any subnet mask you define manually using CIDR notation.
Example: /26, /27, /30, etc.

VISUAL UNDERSTANDING



Network

192.168.1.0/24



Network Part (255.255.255.0)

- Same for all devices.
- Helps devices know they are in the same **network**.

Host Part

- Unique for each device.
- Helps identify devices within the network.

EXAMPLE SCENARIO

IP Address
192.168.1.25



All devices with 192.168.1.x are in the same network.

Subnet Mask
255.255.255.0



They can communicate directly with each other.

Network ID
192.168.1.0



To reach another network, they need a router.



KEY TAKEAWAY:

Subnet mask is the key to understanding **networks**, **subnets**, and **devices**.



Right mask. Right network. Right communication.

12

SUBNETTING PROCESS – STEP BY STEP

Subnetting breaks a network into **smaller pieces**. Follow the steps to do it right!

WHAT IS SUBNETTING?

The process of dividing a large network into smaller, manageable sub-networks (**subnets**).



Reduces Broadcast Traffic
Improves network performance.



Improves Security
Limits access and isolates departments.

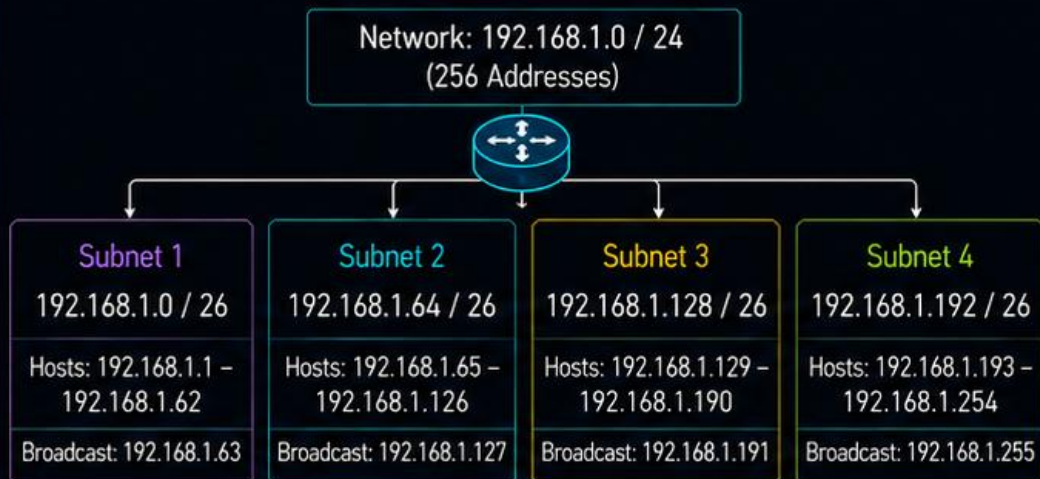


Efficient IP Usage
Helps use IP addresses wisely.



Better Management
Easier to manage and troubleshoot.

EXAMPLE: SUBNETTING A /24 NETWORK INTO 4 SUBNETS



We borrowed **2 bits** from the host part (/24 → /26) to create **4 subnets** ($2^2 = 4$).



Big network. Small subnets.
Better **performance**. Better **control**. ”

SUBNETTING STEPS

1

Identify the Network

Note the given network and prefix.



2

Determine Subnets Needed

Find how many subnets you need.



3

Calculate Bits to Borrow

Find the number of bits to borrow using $2^n \geq$ number of subnets.



4

New Subnet Mask

Add borrowed bits to the network prefix to get new mask.



5

Find Subnets

Calculate network ID, usable range, and broadcast for each subnet.



HOW BITS ARE BORROWED

Example: /24 network → Borrow 2 bits → /26



Borrowing 2 bits from host part gives us **4 subnets** and **62 usable hosts** per subnet.

SUBNET CALCULATION SUMMARY (FOR /26)

Original Network	192.168.1.0 / 24
New Subnet Mask	255.255.255.192 (/26)
Number of Subnets	4
Usable Hosts per Subnet	62
Block Size (Increment)	64
Subnet Networks	0, 64, 128, 192
Broadcast Addresses	63, 127, 191, 255

GENERAL FORMULA

 2^n

Number of Subnets = 2^n
 n = number of bits borrowed

Usable Hosts per Subnet = $2^h - 2$
 h = remaining host bits



Block Size = 256 – Subnet Mask (last octet) or 2^h



New Prefix = Original Prefix + Bits Borrowed



KEY TAKEAWAY:

Subnetting = **Borrow** bits, **create** subnets, **save** addresses, **improve** networks!



Plan **smart**. Subnet **smart**. Manage **better**!

13

VLSM – SUBNETTING LIKE A PRO

VLSM (Variable Length Subnet Mask) lets us create subnets of **different sizes** as per need.

WHAT IS VLSM?

VLSM allows us to divide a network into subnets of **different sizes** based on the number of hosts required.

WHY USE VLSM?



Efficient Use of IPs

No IP addresses are wasted.



Flexible Allocation

Allocate bigger subnets where needed, smaller where not.



Better Performance

Reduces congestion and improves network efficiency.



Scalable Design

Ideal for real-world network planning.

EXAMPLE: COMPANY NETWORK (10.1.0.0/16)

Requirements:

Department	Hosts Needed	Subnet Size (Prefix)
IT Department	200	/24 (254 hosts)
HR Department	100	/25 (126 hosts)
Sales Department	50	/26 (62 hosts)
Guest Wi-Fi	25	/27 (30 hosts)
Network Management	2	/30 (2 hosts)

VLSM Allocation (From 10.1.0.0/16)

Department	Network Address	Prefix	Usable Host Range	Broadcast Address
IT Department	10.1.0.0	/24	10.1.0.1 – 10.1.0.254	10.1.0.255
HR Department	10.1.1.0	/25	10.1.1.1 – 10.1.1.126	10.1.1.127
Sales Department	10.1.1.128	/26	10.1.1.129 – 10.1.1.190	10.1.1.191
Guest Wi-Fi	10.1.1.192	/27	10.1.1.193 – 10.1.1.222	10.1.1.223
Network Management	10.1.1.224	/30	10.1.1.225 – 10.1.1.226	10.1.1.227



Right size for right need.
No waste. All purpose.



HOW VLSM WORKS

1



List all subnets and their host requirements.

2



Sort requirements from largest to smallest.

3



Find the smallest subnet that fits the requirement.

4



Assign subnets one by one without overlapping.

5



Ensure no overlap and all needs are met.



PRO TIP

Always allocate from large to small and move forward. This avoids gaps and overlap.

SUBNET SIZE QUICK REFERENCE

Prefix	/30	/29	/28	/27	/26	/25	/24	/23	/22	/21	/20
Usable Hosts	2	6	14	30	62	126	254	510	1022	2046	4094
Use Case	P2P Links	Tiny Links	Small LAN	Small LAN	Medium LAN	Large LAN	Large LAN	Huge LAN	Huge LAN	Very Large	Very Large

REAL-WORLD USE CASES



Enterprise Networks

Different departments have different needs.



ISP Networks

Customers get different sized subnets.



Hotels / Campuses

Rooms, staff, guests need different sizes.



Data Centers

Each service needs its own subnet.



KEY TAKEAWAY:

VLSM gives **flexibility** and **efficiency**. Right subnet size for right need = **better network!**



Plan **smart**. Allocate **smart**. **Save IPs**. **Improve performance**.

14

SUPERNETTING (CIDR AGGREGATION) – COMBINE AND OPTIMIZE

Supernetting (CIDR Aggregation) combines smaller networks into a bigger network to **reduce routing entries** and **simplify** the routing table.

WHAT IS SUPERNETTING?

Supernetting combines multiple contiguous networks into a single larger network using a **shorter prefix**.



Reduces Routing Table Size

Fewer routes = smaller routing tables.



Improves Performance

Less processing, faster convergence.



Simplifies Network Design

Easier to manage and troubleshoot.



Efficient Use of Resources

Better route summarization for ISPs and enterprise networks.

EXAMPLE: AGGREGATE 4 CONTIGUOUS /24 NETWORKS

Step 1: Given Networks	Step 2: Binary Representation (Last Octet)	Step 3: Find Common Prefix
192.168.0.0 / 24	192 . 168 . 00000000	First 22 bits are common. 22 common bits = /22
192.168.1.0 / 24	192 . 168 . 00000001	
192.168.2.0 / 24	192 . 168 . 00000010	
192.168.3.0 / 24	192 . 168 . 00000011	
		Common prefix = 22 bits
Step 4: Aggregated (Supernet) Network		
192.168.0.0 / 22		



Covers 192.168.0.0 to 192.168.3.255 (4 x /24 networks)



Fewer routes. Smarter network.

Better **performance**. Simplified **routing**.



BEFORE vs AFTER SUPERNETTING

Before (4 Routes)



192.168.0.0 / 24
192.168.1.0 / 24
192.168.2.0 / 24
192.168.3.0 / 24

4 routes
in routing
table

After (1 Route)



192.168.0.0 / 22

1 route
in routing
table

Result: Less routes, faster routing, simpler network.

HOW TO SUPERNET (STEP BY STEP)

- 1 List all networks to aggregate.
- 2 Convert to binary.
- 3 Find the longest common prefix.
- 4 Use that prefix as the new subnet mask.
- 5 The first address becomes the network ID.

QUICK CHECK

- Networks must be contiguous.
- The number of networks to aggregate must be a **power of 2** (2, 4, 8, 16...).
- All networks must have the same prefix length.



MORE EXAMPLES

Networks to Aggregate	Total Networks	New Prefix	Aggregated Network	Covers
10.0.0.0 / 24 – 10.0.1.0 / 24	2	/23	10.0.0.0 / 23	10.0.0.0 – 10.0.1.255
10.0.0.0 / 24 – 10.0.3.0 / 24	4	/22	10.0.0.0 / 22	10.0.0.0 – 10.0.3.255
172.16.0.0 / 24 – 172.16.7.0 / 24	8	/21	172.16.0.0 / 21	172.16.0.0 – 172.16.7.255
192.168.0.0 / 24 – 192.168.15.0 / 24	16	/20	192.168.0.0 / 20	192.168.0.0 – 192.168.15.255

WHERE SUPERNETTING IS USED



ISPs

Advertise summarized routes to the internet.



Enterprise Networks

Reduce internal routing overhead.



Route Summarization

Advertise a single summary route instead of many.



WAN Optimization

Less routes across WAN = better performance.



KEY TAKEAWAY:

Aggregate **smart**. Route **less**. Perform **more**.



Supernetting **saves space** in routing tables and **speeds up** your network.



Think **bigger**. Route **smarter**. Network **better**.

SUBNETTING PRACTICE – SOLVE LIKE A PRO!

Master subnetting with **practice**. Follow the steps, **avoid mistakes**, and **build confidence**.

“ Practice builds **accuracy**.
Accuracy builds **expertise**. ”

THE 5 GOLDEN STEPS

-  **Understand the Requirement**
Identify network, hosts, and goals.
-  **Convert & Calculate**
Convert to binary, find prefix and mask.
-  **Find Subnets**
Determine subnet ID, range, broadcast.
-  **Verify & Validate**
Check host counts and ranges.
-  **Document & Summarize**
Present clearly and double-check.

PRACTICE PROBLEM

You have been assigned the network **172.16.0.0/16**.
You need **6** subnets. Each subnet must support at least **50** usable hosts.
Find the new subnet mask and list the first 2 subnets with details.

STEP-BY-STEP SOLUTION

Step 1: Requirement	- Need 6 subnets → borrow bits - At least 50 usable hosts per subnet → host bits
Step 2: Calculate	- Hosts needed: 50 → find h where $(2^h - 2) \geq 50$ → h = 6 (62 usable) - Borrowed bits for subnets: need 6 subnets → $2^n \geq 6$ → n = 3 ($2^3 = 8$ subnets)
Step 3: New Prefix & Mask	- Original: /16 • Borrow 3 bits → New prefix: /19 - New Subnet Mask: 255.255.224.0
Step 4: Block Size	Block size in 3rd octet = $256 - 224 = 32$

Step 5: Subnets	Subnet	Network ID	Usable Host Range	Broadcast	Usable Hosts
	1	172.16.0.0/19	172.16.0.1 – 172.16.31.254	172.16.31.255	8190
	2	172.16.32.0/19	172.16.32.1 – 172.16.63.254	172.16.63.255	8190

 **FINAL ANSWER:** New Subnet Mask: **255.255.224.0** (/19)
You get **8** subnets (≥ 6 needed), each with **8190** usable hosts (≥ 50 needed).

COMMON MISTAKES TO AVOID

-  **Wrong host calculation**
Forgetting to subtract 2 from 2^h .
-  **Wrong block size**
Use $256 - \text{mask value}$, not mask value.
-  **Incorrect ranges**
Include network ID or broadcast in usable range.
-  **Not enough subnets**
Always ensure $2^n \geq \text{required subnets}$.
-  **Overlapping subnets**
Each subnet must be unique and non-overlapping.

QUICK CHECK TIPS

-  Does the number of subnets meet or exceed the requirement?
-  Does each subnet have enough usable hosts?
-  Are subnet ranges correct and non-overlapping?
-  Are network ID and broadcast correctly identified?
-  Is the final answer clearly summarized?

MORE PRACTICE – TRY THESE!

Problem	Network	Subnets Needed	Usable Hosts / Subnet
1	192.168.0.0/16	10	100
2	10.10.0.0/16	20	200
3	172.20.0.0/16	40	500
4	192.168.100.0/24	4	50
5	10.0.0.0/8	60	1000

Use the **5 Golden Steps** for each problem.

-  Analyze → Convert →
-  Calculate → Verify →
-  Summarize 

SUBNETTING MINDSET

-  **Plan Before You Calculate**
Understand the goal before doing math.
-  **Think in Binary**
Binary is the language of IP.
-  **Double-Check Always**
Small mistakes cause big issues.
-  **Practice Regularly**
More practice = more confidence.

HANDY REFERENCE

Hosts Needed	Host Bits (h)
Up to 2	2 ($2^2 - 2 = 2$)
Up to 6	3 ($2^3 - 2 = 6$)
Up to 14	4 ($2^4 - 2 = 14$)
Up to 30	5 ($2^5 - 2 = 30$)
Up to 62	6 ($2^6 - 2 = 62$)
Up to 126	7 ($2^7 - 2 = 126$)
Up to 254	8 ($2^8 - 2 = 254$)

 **REMEMBER:**
 $2^h - 2$ gives
usable hosts
per subnet.



KEY TAKEAWAY:

Practice makes you **perfect**.

Follow the **steps**. Avoid **mistakes**. Think **bigger**.



Confident today. **Expert** tomorrow!

KEY CONCEPTS



- Subnetting divides a network into smaller, manageable subnets.
- Subnet Mask helps separate **Network Part** and **Host Part**.
- VLSM** allows different size subnets as per requirement.
- Supernetting (CIDR Aggregation) combines smaller networks to reduce routes.
- Goal: Efficient IP usage, better performance, and simplified routing.

COMMON SUBNET MASKS

CIDR	Subnet Mask	Usable Hosts
/8	255.0.0.0	16,777,214
/16	255.255.0.0	65,534
/24	255.255.255.0	254
/25	255.255.255.128	126
/26	255.255.255.192	62
/30	255.255.255.252	2

SUBNET MASK QUICK REFERENCE

Prefix	/30	/29	/27	/26	/25	/24	/23	/22	/21	/20
Usable Hosts	2	6	30	62	126	254	510	1022	2046	4094
Use Case	P2P Links	Tiny Links	Small LAN	Small LAN	Medium LAN	Large LAN	Huge LAN	Huge LAN	Very Large	Very Large

KEY FORMULAS



- Number of Subnets = 2^n (n = number of borrowed bits)
- Usable Hosts per Subnet = $2^h - 2$ (h = remaining host bits)
- Block Size = 256 – Subnet Mask (last octet) or 2^h
- New Prefix = Original Prefix + Bits Borrowed
- $2^h - 2$ gives usable hosts per subnet.

SUBNETTING PROCESS (5 GOLDEN STEPS)

- 1 Understand the Requirement** Identify network, hosts, and goals.
- 2 Convert & Calculate** Convert to binary, find prefix and mask.
- 3 Find Subnets** Determine subnet ID, range, broadcast.
- 4 Verify & Validate** Check host counts and ranges.
- 5 Document & Summarize** Present clearly and double-check.

SUBNETTING MINDSET



- Plan Before You Calculate** – Understand the goal before doing math.
- Think in Binary** – Binary is the language of IP.
- Double-Check Always** – Small mistakes cause big issues.
- Practice Regularly** – More practice = more confidence.

PRACTICE PROBLEM (RECAP)

Given: Network 172.16.0.0/16
Need: 6 subnets, each with at least 50 usable hosts.
Find: New subnet mask and first 2 subnets.

Final Answer:
New Subnet Mask: 255.255.224.0 (/19)
You get 8 subnets (≥ 6 needed), each with 8190 usable hosts (≥ 50 needed).

Subnet	Network ID	Usable Host Range	Broadcast	Usable Hosts
1	172.16.0.0/19	172.16.0.1 – 172.16.31.254	172.16.31.255	8190
2	172.16.32.0/19	172.16.32.1 – 172.16.63.254	172.16.63.255	8190

SUPERNETTING (CIDR AGGREGATION)

- Combine contiguous networks with same prefix.
- Reduces routing table size.
- Improves performance.

Example: Aggregate 4 contiguous /24 networks

192.168.0.0 /24
192.168.1.0 /24
192.168.2.0 /24
192.168.3.0 /24



Aggregate (Supernet)
192.168.0.0 /22
(Covers 192.168.0.0 to 192.168.3.255)

✔ **Benefit:** Fewer routes, faster routing, simpler network.

VLSM (VARIABLE LENGTH SUBNET MASK)

- Allows different size subnets as per need.
- Sort requirements (largest to smallest).
- Allocate using suitable prefix.

Department	Hosts Needed	Subnet (Prefix)	Usable Host Range (Example from 10.1.0.0/16)
IT Department	200	/24 (254)	10.1.0.1 – 10.1.0.254
HR Department	100	/25 (126)	10.1.1.1 – 10.1.1.126
Sales Department	50	/26 (62)	10.1.1.129 – 10.1.1.190
Guest Wi-Fi	25	/27 (30)	10.1.1.192 – 10.1.1.222
Network Mgmt	2	/30 (2)	10.1.1.224 – 10.1.1.226



Key: Right size for right need. No waste. All purpose.

BEFORE vs AFTER SUPERNETTING



Before (4 Routes)
192.168.0.0 /24
192.168.1.0 /24
192.168.2.0 /24
192.168.3.0 /24



After (1 Route)

192.168.0.0 /22

Result: Less routes, faster routing, simpler network.



COMMON MISTAKES TO AVOID

- ❌ **Wrong host calculation** (Forgetting to subtract 2 from 2^n).
- ❌ **Wrong block size** (Using 256 – mask value, not mask value).
- ❌ **Incorrect ranges** (Include network ID or broadcast in usable range).
- ❌ **Not enough subnets** (Always ensure $2^n \geq$ required subnets).
- ❌ **Overlapping subnets** (Each subnet must be unique and non-overlapping).

QUICK CHECK TIPS

- ✔ Do subnets meet/exceed host requirement?
- ✔ Are all ranges correct & non-overlapping?
- ✔ Are network ID & broadcast correct?
- ✔ Are you using right block size?
- ✔ Is the final answer summarized?

HANDY REFERENCE

Hosts Needed	Host Bits (h)	Formula
Up to 2	2	$(2^2 - 2 = 2)$
Up to 6	3	$(2^3 - 2 = 6)$
Up to 14	4	$(2^4 - 2 = 14)$
Up to 30	5	$(2^5 - 2 = 30)$
Up to 62	6	$(2^6 - 2 = 62)$
Up to 126	7	$(2^7 - 2 = 126)$
Up to 254	8	$(2^8 - 2 = 254)$



FINAL TAKEAWAY:



Understand deeply.



Plan smart.



Calculate carefully.



Verify always.



Document clearly.



Practice more. Excel always!